

Demystifying Mental Health Reports Through an LLM-based Approach

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Abstract

Medical diagnoses typically contain strong jargon that is inaccessible to people with lower health literacy. Mental health is one such domain where comprehensibility is exacerbated further. Even with increasing awareness about mental health, support from practitioners or authentic knowledge sources is required to understand diagnostic information. Through this work, we utilize locally trained and finetuned Large Language Models (LLMs) to examine comprehension improvement of mental health diagnostic reports. Through a survey (n=105) with participants over 18 and a demonstration to practicing professionals, we examine the outputs of our AI-driven approach. Our approach offers two key benefits: (1) it simplifies complex psychiatric information for laypersons, providing understandable explanations; (2) it enables doctors to access concise summaries and information via tailored retrieval techniques. We outline this tool's development, explain the methodology, evaluate its potential for information gain and healthcare delivery, and propose directions for future research.

* Authors are advisors of this research work.

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CCS Concepts

• **Information systems** → **Specialized information retrieval**; • **Human-centered computing** → *Interactive systems and tools*; • **Computing methodologies** → **Information extraction**; • **Applied computing** → **Health care information systems**; **Health informatics**.

Keywords

Health Informatics, Information Extraction, Medical Records Informatics

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1 Introduction

With the increased popularization of medical information [20], health information-seeking behavior has increased recently [4]. While health priorities (particularly mental health [24]) have accelerated post-COVID, people still find it challenging to understand health information with medical terminology [13, 14, 18]. In India, this challenge is even more pronounced in mental health, a domain often overlooked due to stigma [29] and limited resources [7].

However, information comprehensibility of mental health reports demands domain knowledge and intricate extraction processes.

Mental health reports are unique in their complexity and sensitivity. Psychiatric data often encompasses nuanced diagnoses, multi-faceted treatment histories, and personal narratives (often including data collected through digital phenotyping [25]). While practitioners can understand the various components of this information collection process, individuals find it challenging. Even with increasing awareness of mental health disorders [30, 33], it can still be overwhelming and challenging to interpret for people, often leading to confusion or disengagement with their care [27]. While reports are comprehensible for doctors, the absence of quickly accessible and well-structured data results in inefficiencies, limiting their ability to deliver timely and personalized care. Addressing these challenges is critical to ensuring that mental health services in India are not only more accessible but also more effective and user-centric.

Through this work, we present a novel solution: an offline generative AI-driven approach to creating structured and accessible mental health information and understandable explanations. By leveraging advanced computing capabilities of Large Language Models (LLMs) and finetuning techniques, we trained models capable of organizing unstructured psychiatric data from discharge summaries into actionable formats. Our work aimed to evaluate how an AI-based approach can improve practitioners' and individuals' information comprehensibility. First, we evaluated the generated comprehensible explanations compared to their counterparts littered with medical terminology using a survey of people aged 18 or older. Second, we evaluated the dashboard with access to concise, tailored summaries of mental health records with practitioners and elicited further recommendations in statistics and trends that might help in their caregiving.

Integrating generative AI in this context can reshape how mental health reports are managed and utilized, indicating the potential for developing a broader system. Beyond the immediate outcome, this approach promptly aligns with India's broader push toward health literacy awareness campaigns and digitizing healthcare. Addressing critical gaps in accessibility and usability contributes to a more equitable, resilient, and user-centered healthcare vision.

In the following sections, we examine the current literature and outline the methodology behind our AI-driven solution, detailing the processing of relevant information and training models to organize and retrieve mental health data effectively. We evaluate the potential impact of this approach on the general Indian adult demographic using a survey and a discussion of its benefits for individuals and doctors alike. Finally, we propose future research directions, emphasizing the need to explore generative AI in healthcare settings further. By demonstrating the feasibility and potential of our approach, this study lays the groundwork for the next generation of digital health innovations in India, with mental health as a critical starting point.

2 Related Work

Mental health care information, specifically digital mental health information, presents distinct challenges that differentiate it from

other areas of healthcare [1]. Over the years, human-centered research has explored digital technology in tackling various aspects of mental health, such as guiding on designing adequate support for mental health [26] or designing interventions that directly improve mental health through bio-feedback mechanisms [11]. While design approaches can help globally, adoption of specialized technology would be limited in the Indian context.

The challenges in presenting mental health information to individuals lie in the need for the right tone, structure, emotional stance, detail, and other factors. Mental health literature [15] highlights that these factors must be considered to make a diagnosis and treatment plan effective. However, psychiatric data is often dense, nuanced, and sensitive, making it difficult for laypersons to interpret and engage with their mental health records even if they are detailed [30, 33]. In the Indian context, doctors spend barely 2 minutes with individual care seekers to explain diagnoses [28], and laypeople find it challenging to understand medical jargon [34]. This leads many individuals to struggle to understand their diagnoses or treatment plans [31], leading to confusion, disengagement, or even mistrust of the healthcare process [27]. This complexity in understanding mental health data motivates the need for more accessible and structured approaches, mainly when dealing with discharge summaries. This work aims to address these challenges by providing a novel approach to organizing and presenting mental health diagnoses.

For doctors, accessing concise and relevant summaries of patient mental health histories is also challenging, especially when digital records do not have accessible filters or retrieval mechanisms. While eHRs may be less prevalent in the Indian context, each medical care facility (including mental healthcare facilities), by law, must provide a *discharge summary*. Practicing psychiatrists also have to perform an initial Mental Status Examination (MSE) [22], detailed through questions from various factors such as demographics, socioeconomic status, mental illness history, and more. When eHRs are lacking, this information takes up a severe overhead in the processing time of caregiving.

LLMs have shown immense potential in addressing challenges related to the organization and retrieval of unstructured healthcare data [23]. Natural language processing (NLP) and generative AI, or LLMs in particular, have been employed to process large volumes of clinical text, converting it into structured and actionable insights [12]. However, as the field of medicine is expansive, nuanced approaches are required to train these models specifically catering to specialized fields. For example, AI-driven solutions have been used to summarize patient histories, retrieve relevant case information, and generate personalized treatment recommendations in fields like oncology [5, 6] and radiology [19, 35]. Another example of the use of generative AI, specifically in the intersection of information retrieval and health records, is the development of an LLM-based clinical note management system to improve clinician productivity by Han et al. [17]. Through this work, we collaborated with certified professionals to understand how this information is presented in discharge summaries and design the retrieval of the same through a finetuned LLM locally.

Despite the success of generative AI in various medical fields, its application in mental health records remains limited due to the unique complexity of psychiatric data. This work aims to fill

this gap by leveraging generative AI trained on domain-specific mental health information, introducing a novel approach to creating structured, accessible mental health records that can be useful for doctors and individuals alike.

3 Understanding Discharge Summaries

A *Discharge Summary Report* is a document completed by healthcare providers when patients leave the hospital, outlining their medical history, current condition, treatments, and discharge instructions. It serves as an important reference for future care. However, in the Indian context, most of these reports have information with heavy medical jargon, and the format varies between institutions. Hence, extracting this information into structured data is time-consuming. While it is easy for clinical psychologists and certified professionals to understand, training LLM or AI models requires detailed concept segmentation and definition of parameters. These parameters vary for each specialization in medicine, and we focused on mental health for this work. With the help of a team of practicing doctors with clinical psychology experience and computer science researchers, we designed a document (named *Case History Taking Template* (CHTT)) to segment various concepts and parameters in case history taking. CHTT contains more detailed information than discharge summaries. The concepts covered in the case history document are categorized as below:

- (1) sociodemographic details
- (2) chief complaints (factors of illness)
- (3) history of present illness (chronologic description of the evolution of symptoms of the current episode)
- (4) past history (all psychiatric illnesses and their course over the patient's lifetime, including family history)
- (5) personal history (chronological account of the patient's personal experiences starting with birth)
- (6) premorbid personality (individualized styles of dealing with the environment that is characteristic to each person prior to the onset of psychiatric disorder)
- (7) mental status examination (the psychiatric equivalent of the physical examination in other medical specialties, entails a comprehensive assessment, inclusive of speech, mood and affect, thought, perception, cognitive functions, judgment, psychomotor activity, and general appearance & behavior)

These categories are further sectioned based on the information appropriate to various aspects of the examined factor. The case history document and other mental health manuals are preprocessed appropriately with context windows and utilized for training on mental health information locally using an open-source pre-trained LLM (*llama 3.3 70b*). With the labeled data from the output of the pre-trained LLM and domain-specific data of the discharge summaries (like attributes and fields of the document), a finetuned LLM is produced for downstream tasks. Finally, we prompted the finetuned LLM for the two desired tasks: structured data extraction and understandable diagnosis explanations. The Design Architecture of the training process used is given in figure 1. The finer details of the data preprocessing, training, and finetuning of the LLM are omitted in the interest of the nature of this submission.

4 Methodology

We utilized respective target populations to evaluate the comprehensibility of explanations generated in our study. The generated explanations of example discharge summaries were evaluated through a survey, where participants aged 18 and older were asked to rate their preference between the generated summaries and the original discharge summaries in terms of detail, clarity, and ease of understanding. Additionally, qualitative feedback was collected to identify aspects of their preferences. The survey responses were analyzed quantitatively, using basic descriptive statistics, and qualitatively, using simple coding of their reasoning for the preferences. This evaluation allowed us to assess whether the AI tool improved a layperson's engagement with mental health diagnoses to provide a more comprehensive experience. The statistics, collective summaries, and analytics derived from the extracted structured data were evaluated with the help of a live demonstration with an FGD of practicing doctors, research scientists, and administrators. Apart from the summaries and statistics shown in the demonstration, we obtained the preferences and expectations of the practitioners regarding other information retrieval.

4.1 Survey

Preference testing between the generated explanation and the original discharge summary was performed in the form of a survey. Through purposive and snowball sampling, the survey was floated through various digital channels like email, WhatsApp, and Instagram. The only limitation imposed on the demographics was that participants had to be aged 18 or older, of Indian origin, and had access to complete a Google form. One hundred five participants completed the survey, with 49 females, 52 males, and 4 preferring not to reveal the gender they identified most.

The survey was designed without explicitly labeling the options while aiming to understand the preferences between the two. Four example discharge summaries (taken from anonymized reports), along with an AI-generated option for each, were presented in the survey. Each example was followed by questions that gauged the participants' preferences between the two options regarding clarity, detail, and ease of understanding. Both options were presented as a choice on either end on a Likert scale. All Likert scale questions had the following description from left to right (1-5): 1: Option 1 is Better than Option 2, 2: Option 1 is Slightly Better than Option 2, 3: Both options are about the Same, 4: Option 2 is Slightly Better than Option 1, 5: Option 2 is Better than Option 1. Detailed descriptive statistics of the survey are presented in 1, and a sample of a discharge summary with its alternative explanation generated by AI is given in figure 2. As we wanted to understand the general preference for diagnosis information presentation, the survey participation was open to all adults who had experience with diagnosis/discharge reports.

4.2 Dashboard Demonstration with FGD

We developed and deployed a local dashboard with limited data samples from the structured data extraction through the LLM model to evaluate the components that benefit the doctors, researchers, and administrators with an FGD (n=6). This dashboard was shown with filters for all the extracted fields and summary generation

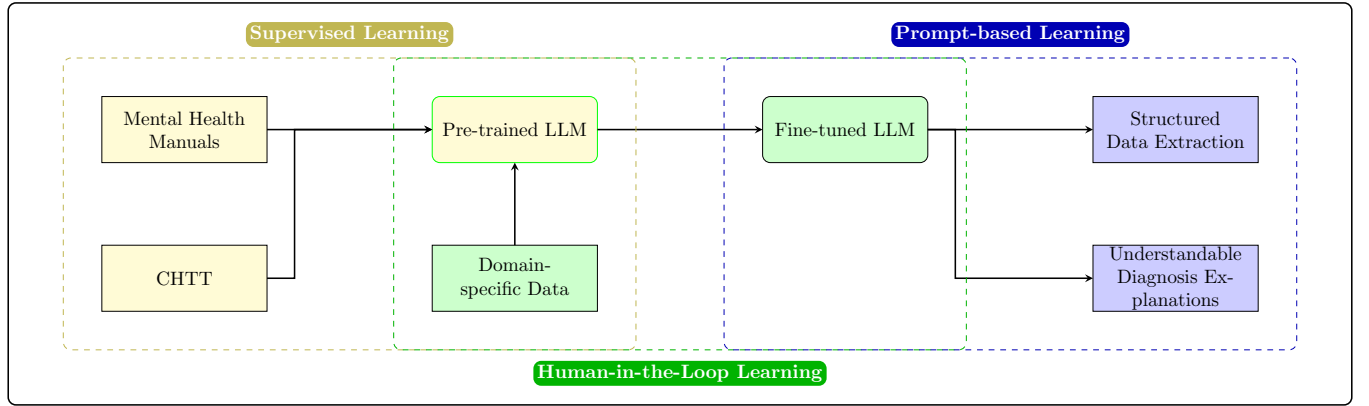


Figure 1: Design Architecture of the training process.

	Total	Average Ratings			Gender			Age	
		Clarity	Detail	Ease of Understanding	Male	Female	Prefer not to say	Range	Mean
Example 1 Option 1	28	20	92	12	16	12	0	19-75	47
Example 1 Option 2	77	85	13	93	36	37	4	18-72	36
Example 2 Option 1	27	18	90	17	17	10	0	19-73	43
Example 2 Option 2	78	87	15	88	35	39	4	18-75	35
Example 3 Option 1	32	32	91	29	14	17	1	18-75	42
Example 3 Option 2	73	73	14	76	38	32	3	19-73	38
Example 4 Option 1	49	36	94	41	28	20	1	19-75	41
Example 4 Option 2	56	69	11	64	24	29	3	18-73	38

Table 1: Descriptive staistics of the survey outcome.

options for patient demographics, symptoms, medication, comorbidities, MSE, history of mental illness, and discharge outcome. Post the demonstrations, the doctors tried using the tool and provided further feedback on presentation, usability, and utility. The doctors also provided recommendations for other information to be presented in the summaries and charts to find health trends across demographics and temporality quickly. A sample screenshot of a couple of screens from the demonstration is presented in 3.

5 Findings and Discussion

5.1 Layperson Evaluations and Understanding Preferences

The layperson evaluations were analyzed using the survey results. Over 70 percent (76/105) of the participants preferred the explanations over the original discharge summary for three examples. The critical difference in the increased number of participants choosing the original discharge summary for the fourth example could be attributed to its short length compared to the other three. There were 68 open-ended responses, and after filtering repetitions and those that did not reason their choice or comparison, 48 responses were identified. Of the participants who chose the original discharge summaries, a few also indicated that they felt these were for medical professionals and should not be given directly to the care seekers. Overall, while participants preferred the generated explanation's

clarity and ease of understanding to be better on average, they favored the original discharge summary for detail. This preference indicates that even with specialized prompting using a finetuned AI model, more effort is required to identify the factors influencing the summary's quality, including its detail, tone, intended target, and many more, which we aim to take up in the follow-up study.

When analyzing the open-ended questions, several descriptions were consistently used to describe both options and the reasoning for their choice. For the original discharge summary, *too medical, better detail, better for the doctor, second opinions, professional detachment, strict instructions with actionable and painful to read* were used as it was evident that while more detailed, the verbose nature made it hard to get through it without prior medical knowledge.

"Understanding my disease is not an option for me because I won't be matured enough to understand and act. Instruction with actionables in a strict tone would help me to shut up and follow." - F 27

For the generated explanation, *easy to read, coherent story, easy to apprehend, not cluttered, personalized, interpretable, brief with clarity, and better for patients* were used, as it indicated that people generally preferred medical information with as little jargon as possible which is reinforced by the findings of prior work [3].

"I suggest option 2 over 1, simply because it was easy to understand and brief. Yes, option 1 was much detailed,

You have been facing challenges over the past two years, including irritability, aggression, social withdrawal, muttering, unusual hand movements, and difficulty expressing emotions. These symptoms started suddenly, possibly triggered by distressing news of a celebrity's death, and significantly impacted your daily life and studies. Medical tests were conducted and found that your vitamin B12 levels were low, for which you were given Neurobion Forte. A small cyst was also detected, but it didn't require immediate treatment. Initial medications like Olanzapine didn't help much, so your treatment was adjusted. You were started on Aripiprazole, which was gradually increased to 25 mg daily, along with Clonazepam 0.25 mg at night and Valproate 200 mg twice daily. Over time, these treatments helped reduce your symptoms, including hallucinations and aggression. You also began performing better in daily tasks, especially at home. At discharge, your condition had improved, and you were prescribed Aripiprazole 25 mg daily, Divalproex 250 mg twice daily, Clonazepam 0.25 mg at night, Neurobion Forte daily, and THP BD daily. You are advised to continue these medications and attend follow-up appointments in the psychiatric outpatient department, along with a psychological assessment, to monitor your progress and adjust your treatment as needed.

Figure 2: Original Discharge Summary (left) and AI-Generated Explanation (right).

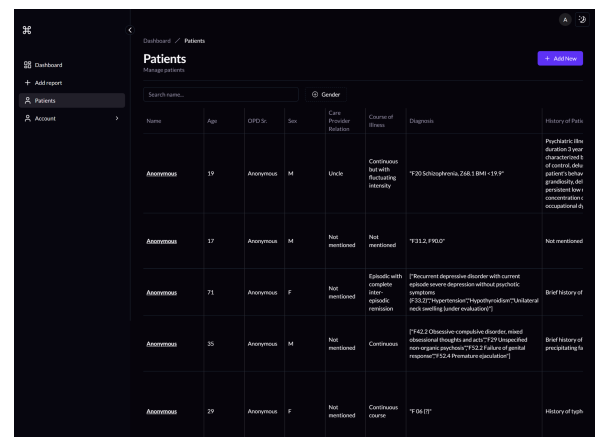


Figure 3: Primary Dashboard Screen (left) and Scrollable/Searchable Database Table (right).

and the observation indicates a possible gap in their mental health literacy. Such hypothesis is further reinforced by prior research on how health literacy can affect a person's ability to seek care [15, 32] and how more likely they are susceptible to more harm due to lack of seeking care, especially for mental health [16, 21].

While the evidence indicates a need for efficient tools for understandable explanations, it is also important to consider how the information is presented and if all the necessary information is captured in the generated explanation. Our results indicate the need to generate unique summarizations while maintaining the information context and relevance, catering to doctors, care seekers, and family members separately.

5.2 Doctor and Administrative Evaluations

Doctors expressed interest in the generated summaries, particularly appreciating the statistical insights provided, such as trends in patient demographics, the prevalence of mental health disorders, and treatment outcomes. They appreciated the ability to quickly access information about common disorders, co-occurring conditions, and symptoms helpful in making more informed decisions. The summaries, which highlighted key trends like age, gender, ethnicity distributions, and the types of treatments being administered, offered valuable context for understanding patient profiles and the potential for tailoring care plans. Of the recommendations, one of the doctors expressed the need to get statistics on seasonal changes.

"I think having information on how the prevalence of mental health disorders changes during different seasons based on weather and festivals could help us understand confounding factors in the patients' environments that affect their conditions."

The summaries and trends demonstrated to and requested by the doctors fall into the following categories: patient demographics and trends, prevalence of mental health disorders, treatment and intervention data, clinical outcomes and effectiveness, follow-up and utilization of services, access to care and barriers, special population data (Percentage of patients from special demographics), and mental health and chronic conditions cooccurrence data. Among other suggestions that were not within the scope of the current tool's capabilities, one was a more futuristic vision for the tool.

"I think these collective summaries are great. However, if can collect data from returning patients and integrate them with their previous discharge summaries, we should be able to examine what factors led to their relapse. This could help us form more evidence-based practices."

The results of this evaluation indicate that this tool has the potential to help both productivity and the development of evidence-based practices. This observation can be further corroborated by prior research on utilizing structured mental health information for developing evidence-based practices and policy-making [9]. Research from other domains, such as knowledge discovery and data mining [36], have also indicated the importance of structured mental health information for developing evidence-based practices. Another potential scope for this tool could be the development of a broader health system directory for analytics, which can be seen from prior work in big data analytics, including healthcare[2, 8], with research on developing architectures for increasing the reliability of healthcare systems [10]. As the tool evaluated was a proof of concept to understand its utility, we aim to develop better deployable systems in the ongoing work

6 Future Work

In this early stage work, we leveraged localized LLMs (*llama 3.3 70b*) to transform unstructured mental health reports into structured data and developed easily understandable summaries free of medical jargon. Our evaluation of generated summaries with laypersons demonstrated that over 70 percent preferred these simplified summaries over traditional discharge summaries, even though they

preferred the detail of the original summary, as they provided more simple and understandable explanations of the diagnosis. Additionally, the evaluation with doctors and administrators revealed that the structured data could enhance administrative productivity and support evidence-based care by helping clinicians stay informed on patient histories and treatment trends.

A limitation of this work is that we trained and evaluated output from a single LLM model (*llama 3.3 70b*). This might limit the findings specific to this open-source model, and we would like to test it with multiple open-source models in future work. The next step is to explore the optimal presentation of these summaries with care seekers with lived experiences or histories of mental health disorders. We aim to integrate insights from mental health literature that highlight specific metrics for evaluating how individuals with mental health disorders perceive speech, text, and any method of interaction about their diagnosis. Including this perspective will guide the development of tailored presentation methods to ensure the summaries are accessible and effective for diverse patient populations. By refining this approach, we aim to enhance an individual's understanding and engagement further, paving the way for more personalized and inclusive mental health care.

Additionally, in the evaluation with the doctors, they explained their ideas on integrating more data for returning patients. They proposed that treatment completion rates and the effectiveness of different therapies be included while developing the overall system as a helpful starting point for evaluating patient progress. They also discussed the impact of incorporating additional data, such as follow-up care utilization and crisis intervention usage, in enhancing the summaries. Understanding how patients engage with aftercare or the frequency of telehealth usage might provide deeper insights into the effectiveness of ongoing treatments. While doctors generally found the summaries to be an improvement over traditional records, they indicated that further refinement, including more granular data on treatment adherence and long-term recovery outcomes, is required to make the tool more valuable in supporting clinical decision-making. Incorporating all these requests could help build a more robust system that can help doctors be more effective quickly while increasing the number of care seekers who could take their help.

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